

BOOLEAN GRAM CLOSURE AS A FINITE DYNAMICAL SYSTEM

ANONYMOUS

ABSTRACT. Let $\mathbb{B} = \{0, 1\}$ with Boolean addition and multiplication, and iterate $\Gamma_n(A) = AA^\top$ on the $n \times n$ Boolean matrices. We give the complete finite functional graph. If $G = AA^\top$, then $\Gamma_n^t(A) = G^{2^{t-1}}$ for $t \geq 1$. Thus every orbit ends at the looped clique completion of the active components of the row-intersection graph. The recurrent states are precisely partial equivalence relations, all fixed; a nonfixed source has depth $1 + \lceil \log_2 D(G) \rceil$, with a zero logarithmic term for $D(G) \leq 1$. The sharp carrier height is zero at $n = 1$ and $1 + \lceil \log_2(n-1) \rceil$ for $n \geq 2$. There are B_{n+1} fixed states. Independently, for every prescribed target we count the complete ordered-column fibre by inclusion–exclusion over its loop and edge requirements. The image criterion is a loop-sensitive edge-clique-cover bound. Boolean powers, transitive closure, graph intersection representations, clique covers, and symmetric Boolean factorization are treated as prior mechanisms; the note remains on external hold.

1. THE LITERAL MAP AND ITS GRAPH FACTOR

Write $\mathbb{B}^{n \times n}$ for the set of zero–one matrices, with

$$(PQ)_{ij} = \bigvee_{r=1}^n (P_{ir} \wedge Q_{rj}).$$

Our self-map is

$$(1) \quad \Gamma_n : \mathbb{B}^{n \times n} \longrightarrow \mathbb{B}^{n \times n}, \quad \Gamma_n(A) = AA^\top.$$

If $R_i = \{r : A_{ir} = 1\}$ is row i regarded as a subset of $[n]$, then

$$(2) \quad \Gamma_n(A)_{ij} = 1 \iff R_i \cap R_j \neq \emptyset.$$

Hence $G = \Gamma_n(A)$ is a symmetric relation. A vertex i has a loop exactly when R_i is nonempty, and every off-diagonal edge has loops at both ends. We call these vertices *active*. On the active vertices, G is an ordinary undirected graph with a loop at each vertex; inactive vertices are unlooped and isolated.

The interpretation (2) is the classical set-intersection representation of a graph. Its equivalence with an edge clique cover goes back at least to Szpilrajn–Marczewski and Erdős–Goodman–Pósa [2, 5]. Boolean matrix powers and transitive-closure algorithms are also classical [4, 6]. More specifically, Fitting records the monotone chain $(AA^\top) \leq (AA^\top)^2 \leq \dots$ and examples with arbitrarily many strict inclusions as dimension grows [3, Theorem 8 and Example 9]. Exact symmetric Boolean factorization $M = WW^\top$ and its clique-cover and hypergraph interpretations are explicit in recent algorithmic work [1]. These mechanisms are background here. Our scope is the exact functional graph of the literal self-map (1), paired with all of its finite ordered-column fibres; no priority claim is made.

For a finite self-map, $\text{depth}(A)$ is the least $d \geq 0$ for which $\Gamma_n^d(A)$ is recurrent. Put $D(G) = 0$ when G has no active edge between distinct vertices; otherwise $D(G)$ is the largest diameter of an active connected component.

Theorem 1 (complete temporal classification). *Let $A \in \mathbb{B}^{n \times n}$ and $G = AA^\top$.*

(i) *For every $t \geq 1$,*

$$(3) \quad \Gamma_n^t(A) = G^{2^{t-1}}.$$

The endpoint is obtained by replacing every active connected component of G by a fully looped clique and leaving every inactive vertex unlooped and isolated.

(ii) *The recurrent states are exactly the partial equivalence relations: disjoint unions of fully looped cliques together with unlooped isolated vertices. Every recurrent state is fixed. Their number is*

$$(4) \quad |\text{Fix}(\Gamma_n)| = \sum_{k=0}^n \binom{n}{k} B_k = B_{n+1},$$

where B_k is the k th Bell number. Consequently the Artin–Mazur zeta function is

$$(5) \quad \zeta_{\Gamma_n}(z) = (1 - z)^{-B_{n+1}}.$$

(iii) The exact source-dependent clock is

$$(6) \quad \text{depth}(A) = \begin{cases} 0, & A \text{ is a partial equivalence relation,} \\ 1 + \lceil \log_2 D(G) \rceil, & \text{otherwise,} \end{cases}$$

where the ceiling term is declared to be zero for $D(G) \leq 1$. The sharp height of the whole carrier is

$$(7) \quad h_n = \max_A \text{depth}(A) = \begin{cases} 0, & n = 1, \\ 1 + \lceil \log_2(n-1) \rceil, & n \geq 2. \end{cases}$$

Proof. The relation G is symmetric, so $\Gamma_n(G) = GG^\top = G^2$. Every Boolean power of G is symmetric; induction therefore gives

$$\Gamma_n(G^{2^s}) = G^{2^s}(G^{2^s})^\top = G^{2^{s+1}},$$

which proves (3).

For active vertices i, j , the entry $(G^r)_{ij}$ is one precisely when there is a walk of length r from i to j . Loops pad a shorter path to any larger length, so this is equivalent to graph distance at most r . Thus $G^{2^{t-1}}$ completes every active component exactly when $2^{t-1} \geq D(G)$, and its stable value is the asserted partial equivalence relation. This proves the endpoint and (6), including $D(G) = 0, 1$.

If $A = AA^\top$, then A is symmetric, every nonempty row has its diagonal entry, and $A = A^2$ is transitive. Its active components are therefore fully looped cliques. Conversely, such a partial equivalence relation A is symmetric and idempotent, hence $AA^\top = A^2 = A$. Every orbit reaches one of these states, so there can be no other recurrent state. To count them, choose the k active vertices and partition them into clique components. This gives the sum in (4); adjoining a distinguished element to the inactive vertices is the standard bijection with partitions of an $(n+1)$ -element set. Every periodic point is fixed, so $|\text{Fix}(\Gamma_n^r)| = B_{n+1}$ for every $r \geq 1$, and the definition of the Artin–Mazur zeta function gives (5).

Finally, $D(G) \leq n-1$, proving the upper bound in (7). For $n \geq 2$, label the edges of the path $1-2-\dots-n$ by $1, \dots, n-1$ and let row i of A contain the labels of the path edges incident with i ; leave column n empty. Then $AA^\top$ is the fully looped path, of diameter $n-1$, and A is not fixed. Its depth is the claimed bound. At $n=1$, both Boolean matrices are fixed. $\square$

2. EVERY-TARGET ORDERED-COLUMN FIBRES

We now solve the one-step inverse problem without assuming that the target lies in the image. For a Boolean relation H on $[n]$, define

$$\mathcal{C}(H) = \{C \subseteq [n] : H_{ij} = 1 \text{ for every } i, j \in C\}.$$

Thus $\emptyset \in \mathcal{C}(H)$, and every nonempty member is a fully looped clique of H . When H is symmetric and each edge has both endpoint loops, let

$$(8) \quad E^*(H) = \{\{i\} : H_{ii} = 1\} \cup \{\{i, j\} : i < j, H_{ij} = 1\}.$$

For $S \subseteq E^*(H)$ put

$$(9) \quad c_H(S) = \#\{C \in \mathcal{C}(H) : e \not\subseteq C \text{ for every } e \in S\}.$$

Theorem 2 (complete target fibre). *If H is nonsymmetric, or if some entry $H_{ij} = 1$ has $H_{ii}H_{jj} = 0$, then $\Gamma_n^{-1}(H) = \emptyset$. For every remaining target,*

$$(10) \quad |\Gamma_n^{-1}(H)| = \sum_{S \subseteq E^*(H)} (-1)^{|S|} c_H(S)^n.$$

Moreover, H lies in the image exactly when the atoms $E^*(H)$ can be covered by at most n members of $\mathcal{C}(H) \setminus \{\emptyset\}$, where C covers e when $e \subseteq C$.

Proof. The necessary symmetry and loop conditions follow immediately from (2). Suppose they hold and write C_r for the support of column r of a prospective source matrix A . Boolean multiplication gives the exact union identity

$$(11) \quad AA^\top = \bigcup_{r=1}^n C_r \times C_r.$$

No zero of H is changed to one exactly when each $C_r \in \mathcal{C}(H)$. Every one of H is obtained exactly when, for each atom $e \in E^*(H)$, at least one chosen C_r contains e .

For a fixed set S of atoms, there are $c_H(S)^n$ ordered column sequences that miss every atom of S . Inclusion–exclusion over the events “atom e is missed by all columns” gives (10). The columns are labelled,

so their order matters; equal column supports are allowed, and the empty support is an allowed padding column. Consequently such a sequence exists precisely when at most n nonempty allowed cliques cover all atoms. $\square$

Ignoring loops, the last statement specializes to the classical intersection-number/edge-clique-cover criterion [2]. The singleton atoms in (8) are essential: an isolated looped vertex still needs a nonempty singleton column support. Compatibility alone is not sufficient. For example, the fully looped $K_{2,3}$ target at $n = 5$ has six edges, every allowed nontrivial clique is an edge, and hence its fibre is zero.

Some small fibres illustrate the ordered convention. Let I_3 and J_3 be the identity and all-one relations, and let P_3^o be the three-vertex path with all three loops. Formula (10) gives

target H	0_3	I_3	P_3^o	J_3
$ \Gamma_3^{-1}(H) $	1	6	30	175

For 0_n , all columns must be empty, so the fibre is one. For an isolated single loop and $n - 1$ inactive vertices, exactly a nonempty subset of the n labelled columns may equal that singleton, giving $2^n - 1$. These examples also expose the zero target, isolated-loop, repeated-column, and empty-column boundaries.

3. EXACT FINITE CONTROLS AND SCOPE

A standalone standard-library verifier exhausts every source and every target through $n = 4$. It compares literal fibres with (10), checks the cover criterion, temporal identity, exact diameter clock, fixed census, and sharp path witnesses; the latter are also replayed through $n = 64$. Its frozen summaries are

n	states	image	fixed	max depth	max fibre
1	2	2	2	0	1
2	16	5	5	1	7
3	512	18	15	2	175
4	65,536	113	52	3	17,887

The same program explicitly exercises $n = 1$, the zero matrix, the $D(G) \leq 1$ convention, isolated loops, unlooped incident-edge targets, and empty columns. Finite computation is evidence against transcription or boundary errors, not a substitute for the all-parameter proofs above.

The claims deliberately stop at this exact package. Boolean semiring algebra, graph squaring and component closure, Bell enumeration, intersection representations, edge clique covers, and inclusion–exclusion are not claimed contributions. In particular, Theorem 2 does not solve the optimization or reconstruction variants of symmetric Boolean factorization. This document is an anonymous author-side Round-0 record with external status `HOLD_EXTERNAL`; it makes no submission, release, novelty, or priority assertion.

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